

JITENDRA VENKAT NALLA

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SUMMARY

Laser Applications Engineer with extensive experience in laser ablation, surface texturing, marking, micro-hole drilling, and micro-welding. In this role, I work closely with customers to understand their challenges and requirements. My responsibility is to develop and optimise processes that meet specified cycle times and tolerances. This involves a deep understanding of laser, materials, laser-material interaction, effective heat management, and a structured, systematic approach to trials, ensuring the final process is stable, repeatable, and ready for production deployment.

TECHNICAL SKILLS

- ✓ R&D for advanced laser applications and its process development.
- ✓ Laser fast micro hole drilling, cold ablation, surface texturing & welding.
- ✓ Commissioning laser systems, beam alignment for both fibre and conventional beam delivery.
- ✓ Expertise in Femto-second - Nano-second (UV – IR) lasers for micro drilling and laser ablation.
- ✓ Pulse modulating on single-mode & multi-mode lasers for welding & single-shot drilling.
- ✓ Calibrating 2 & 3 axis Scanlab scanners and F-theta lenses.
- ✓ Calibrating Optical & Digital microscopes.
- ✓ Worked with multi-axis machines, multi-axis stages and rotary controllers.
- ✓ Designing experiments, designing laser toolpaths and performing trials.
- ✓ Metallurgy of samples, analysing results & writing technical reports.
- ✓ Ability to read/interpret engineering & electrical drawings.
- ✓ Proficient in laserDESK, DMC, WinMark pro, Siemens NX, Heidenhain programming.

WORK EXPERIENCE

- **Laser applications engineer - Winbro Group Tech, UK.** **(April 2023 – Present)**
 - ✓ At Winbro, I have Commissioned laser systems and aligned ultra-short pulse lasers, single mode and multimode laser beams.
 - ✓ Developed a state-of-the-art ultrafast drilling process with speeds up to 3200 holes/second (single shot drilling).
 - ✓ Helped developing a process for drilling holes ranging from 12 – 200 microns for SOFC, hydrogen fuel cells and various other micro hole drilling projects using nanosecond lasers.
 - ✓ Led process development for laser ablation on RR & MTU turbine blades using IR femtosecond laser, achieving sub-10 microns surface roughness.
 - ✓ Developed drilling process for polymers and aluminium silicon-carbide.
 - ✓ Helped in Designing and Setting up the UK Winbro laser lab with floating optical tables, free space optics, 3 axis scanner, inline camera, beam expander and liner stages.
 - ✓ Established a comprehensive database and reports for laser welding, optimising parameters for various materials & thicknesses, for both similar and dissimilar materials.
 - ✓ Utilized Laser Weld Monitoring (LMW) by Precitec for laser weld quality analysis.
- **Supervisor - Raj Constructions, India. (Family Business).** **(Nov 2020 - July 2021)**

EDUCATION

- ✓ **MSc Advanced Manufacturing Engineering and Management.** **(2021 – 2023)**
Loughborough University, Loughborough, UK. (Merit)